Individual Assignment 5

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1. Set up

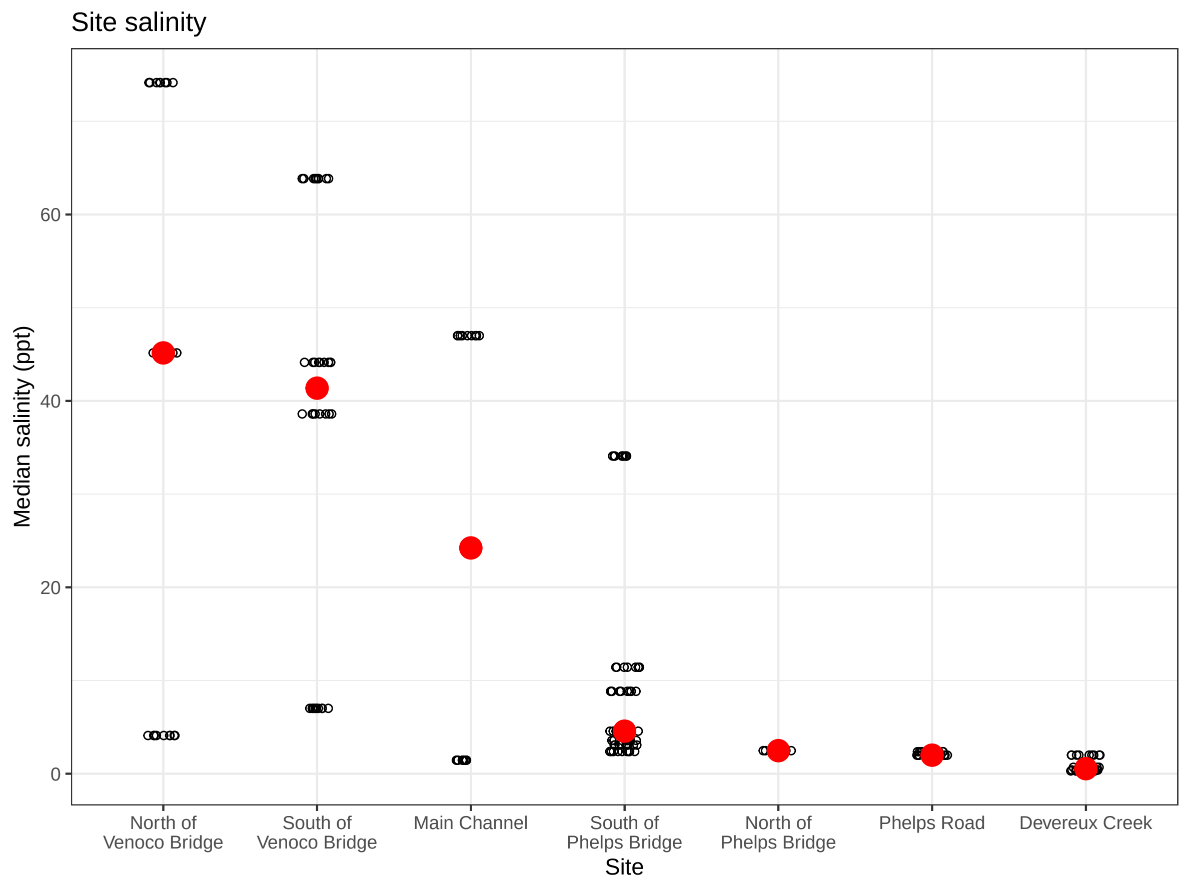
Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

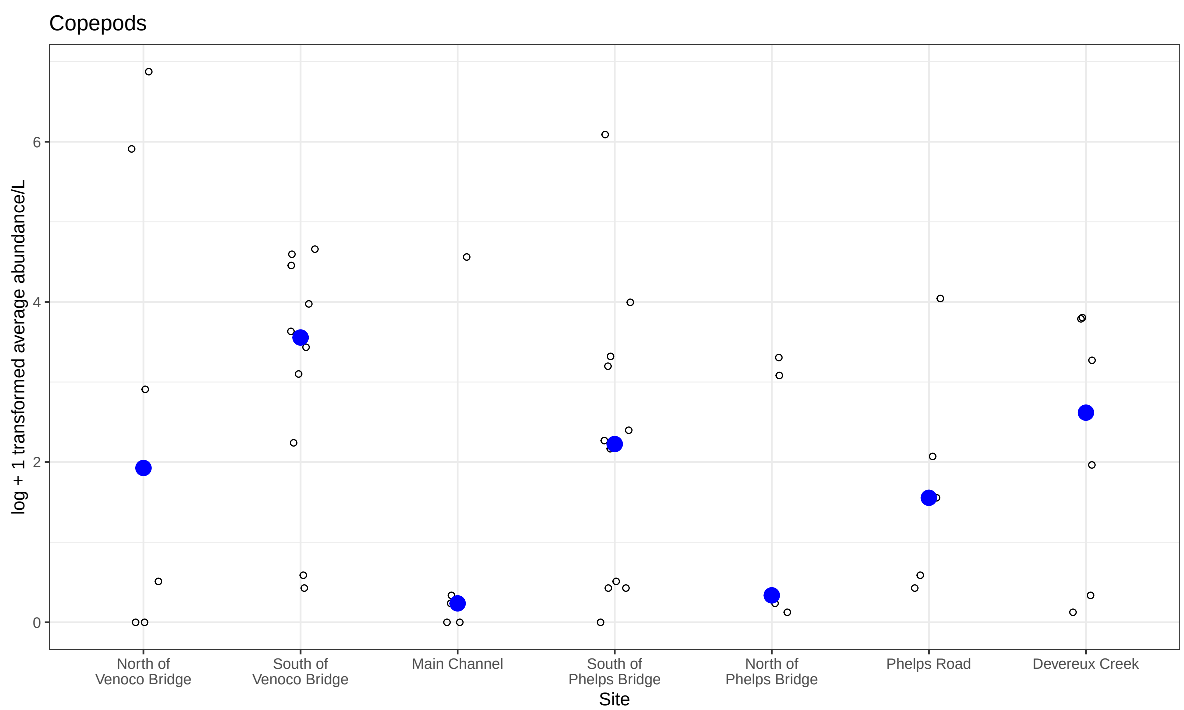
Importing other graphs

```
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()

# displaying object
salinity
```




```r
# base layer
copepods_plot <- ggplot(data = copepods_df,
                        # x-axis: site
                        # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
```

```r
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()

# displaying object
copepods_plot
```



Question 1

A: Planning

- x axis: Site
- y axis: log transformed Ostracod abundance
- geometry abundance: geom_jitter()
- geometry median: stat_summary()

B: Wrangle Data

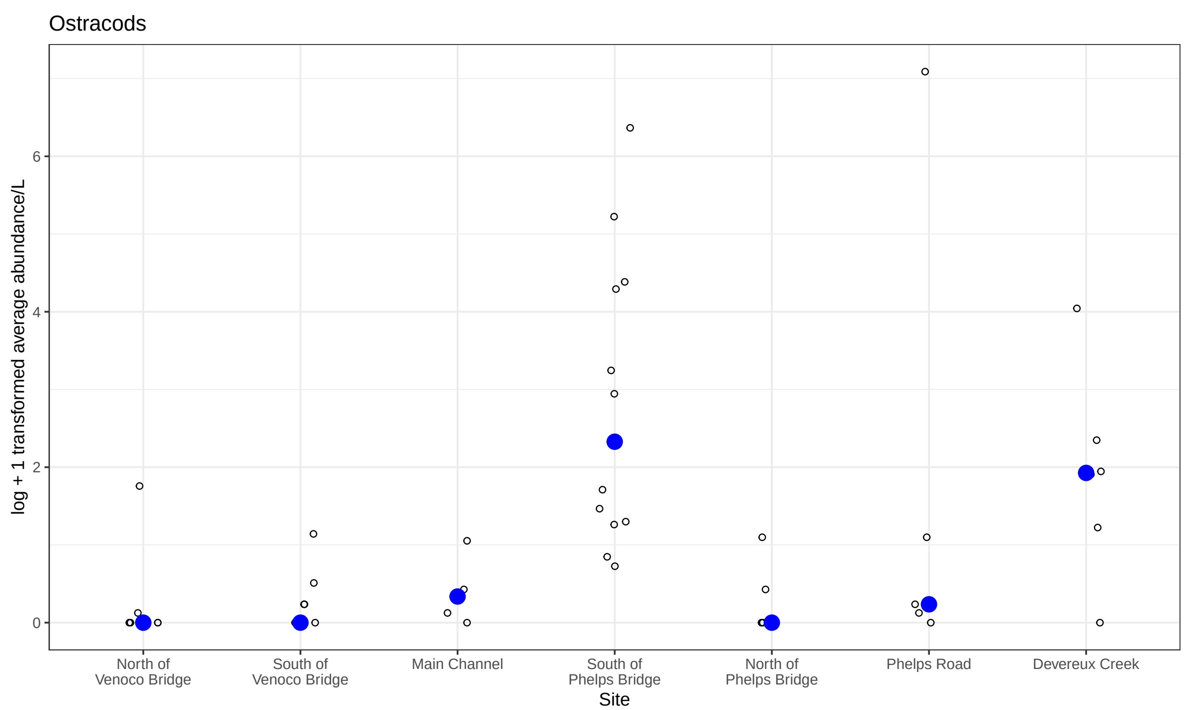
```r
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA     NA      NA
 2 2022-05-11_NDC  2022-05-11 00:00:00 NDC   ostr~   0       9.11   0.398   9.4
 3 2022-08-12_NEC  2022-08-12 00:00:00 NEC   ostr~   0       7.37  NA      NA
 4 2020-01-21_NDC  2020-01-21 00:00:00 NDC   ostr~   2.4     6.53   0.3     7.6
 5 2023-03-03_NDC  2023-03-03 00:00:00 NDC   ostr~   5.73    7.45   0.688   5.64
 6 2023-09-12_NVBR 2023-09-12 00:00:00 NVBR  ostr~   0       9.14  NA      20.3
 7 2022-01-24_NEC  2022-01-24 00:00:00 NEC   ostr~   0.267  NA     NA      NA
 8 2023-02-28_NPB  2023-02-28 00:00:00 NPB   ostr~   2.53   NA     NA      NA
 9 2022-08-12_NVBR 2022-08-12 00:00:00 NVBR  ostr~   0      NA     NA      NA
10 2023-05-04_NMC  2023-05-04 00:00:00 NMC   ostr~   1.87    8.32  NA      10.3
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code Figure

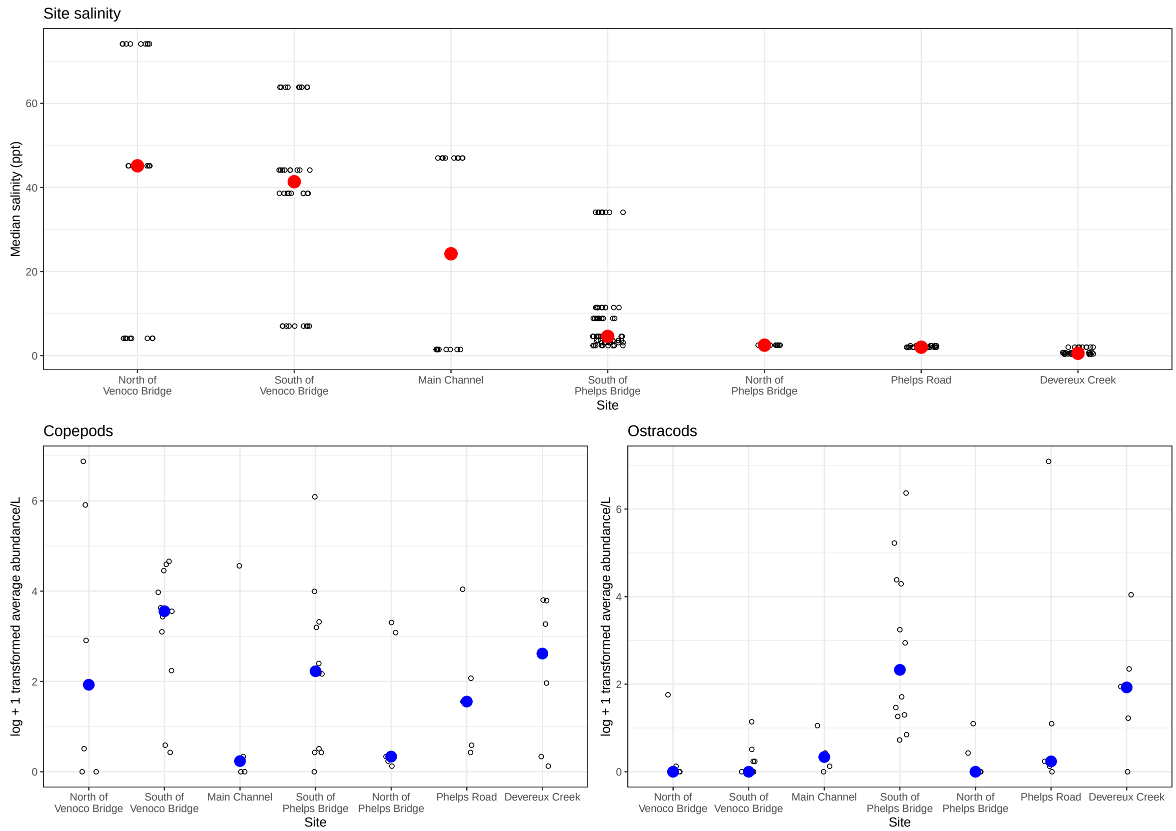
```r
# base layer
ostracods_plot <- ggplot(data = ostracod,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
```

```
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracods_plot
```



D: Create multipanel figure

```
#combining plots
salinity/(copepods_plot+ostracods_plot)
```



E: Analysis

- The differences in sites between copepods and ostracod abundance are mainly at the South of Phelps Bridge as ostracods are the most abundant here while copepods are more abundant South of Venoco Bridge.
- Ostracod abundance may not be related to site-level differences as there are numerous occassions of median salinity being relatively low at places like Phelps Bridge or Phelps road, yet South of Phelps Bridge sites have high abundance while the North side doesn't. Additionally,low abundance sites like North of Phelps have the same abundance as high salinity sites like Venoco Bridge.

Question 2

A: Planning

- x axis: salinity
- y axis: log transformed abundance
- color: taxon
- geometry: geom_point()
- facets: facet_Wrap(~taxon, scales="free")

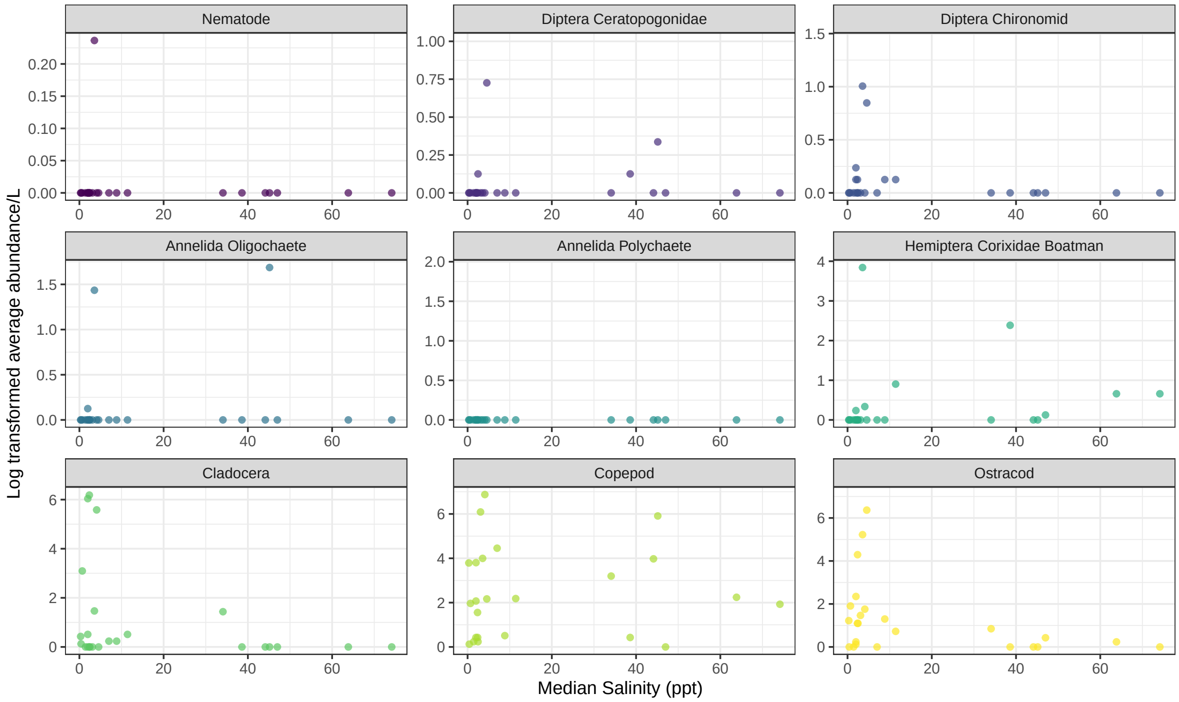
B: Wrangle Data

```
# creating new clean object from aquatic inverts
ab_salinity <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
         # convert taxon to title
         taxon = to_any_case(taxon, case = "title"),
         # order taxon levels by abudance
         taxon = fct_reorder(taxon, ave_lit, .fun="max"))

#display 10 random rows
slice_sample(ab_salinity, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <fct>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-05-11_NDC  2022-05-11 00:00:00 NDC   Ostr~      0    9.11   0.398   9.4
 2 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR  Nema~      0    7.34   4.1     6.1
 3 2021-04-15_NPB  2021-04-15 00:00:00 NPB   Ostr~    185.   8.09   3.57    8.24
 4 2023-02-12_NEC  2023-02-12 00:00:00 NEC   Anne~      0   NA     NA      NA
 5 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  Dipt~      0   NA     NA      NA
 6 2023-06-06_NPB1 2023-06-06 00:00:00 NPB1  Hemi~      0    7.48  NA       2.38
 7 2022-02-21_NPB  2022-02-21 00:00:00 NPB   Anne~      0    7.39  34.1     0.34
 8 2023-02-12_NEC  2023-02-12 00:00:00 NEC   Anne~      0   NA     NA      NA
 9 2022-05-12_NPB2 2022-05-12 00:00:00 NPB2  Clad~    494.  NA     NA      NA
10 2022-11-18_NVBR 2022-11-18 00:00:00 NVBR  Anne~      0   NA     NA      NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

C: Code figure

```
# base layer
ggplot(data = ab_salinity,
       # x-axis: site
       # # y-axis: mean abundance/L
                   mapping = aes(x = med_sal,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # layer 1: points to represent observations
  geom_point(alpha = 0.7) +
  #facet wrap for taxon
  facet_wrap(~ taxon,
             scales = "free") +
  # use color palette
  scale_color_viridis_d() +
  # labels
  labs( x = "Median Salinity (ppt)",
        y = "Log transformed average abundance/L") +
  # theme
  theme_bw()+
  # remove legend from theme
  theme(legend.position = "none")
```



D: Analysis

- The differences between taxa in relationships between abundance and salinity is that the higher the salinity, the lower the average abundance. However, this isn't the case with all species as Copepods are abundant at high and low salinities, similar to Hemiptera as both show up to 2.5 log average abundance/L in higher salinities. On the other hand, species like Annelida Oligochaete, along with the majority of the species, follow an exponential decrease as salinity increases. This includes Cladocera and Diptera Chrionomid to name a few.